

ADVANCED QUANTUM MECHANICS

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Conservation of angular momentum

The angular momentum operator is, written explicitly with all 3 components

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \begin{pmatrix} \hat{y}\hat{p}_z - \hat{z}\hat{p}_y \\ \hat{z}\hat{p}_x - \hat{x}\hat{p}_z \\ \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \end{pmatrix},$$

or in Levi-Civita notation where

$$\varepsilon_{ijk} = \begin{cases} +1, & \text{if } (i, j, k) \text{ is an even permutation of } (x, y, z), \\ -1, & \text{if } (i, j, k) \text{ is an odd permutation of } (x, y, z), \\ 0, & \text{if any two indices are equal.} \end{cases}$$

it can be written as $\hat{L}_i = \sum_{j,k=x,y,z} \varepsilon_{ijk} \hat{x}_j \hat{p}_k$. We have the usual $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij} \mathbb{1}$ and we label the indices i, j, k by x, y, z for compactness.

(i) Prove the following commutations

$$[\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k, \quad [\hat{L}_i, \hat{x}_j] = i\hbar \varepsilon_{ijk} \hat{x}_k, \quad [\hat{L}_i, \hat{p}_j] = i\hbar \varepsilon_{ijk} \hat{p}_k.$$

(ii) Using the rules found above, prove that

$$[\hat{L}_i, V(\hat{r})] = 0.$$

(iii) Prove that

$$[\hat{L}_i, \hat{\mathbf{p}}^2] = 0,$$

Conclusion. For any central potential $V(\hat{r})$ (including the Coulomb potential), the Hamiltonian commutes with all $\hat{L}_i$, hence with $\hat{L}_z$ and with $\hat{\mathbf{L}}^2$:

$$[\hat{H}, \hat{\mathbf{L}}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0.$$

Transition matrix elements and selection rules

We will explore the electric dipole moment operator for an electron in the hydrogen atom.

(i) Starting from the definition of the dipole moment operator for a point charge distribution (for electron and proton),

$$\hat{\mathbf{d}} = \sum_i q_i \hat{\mathbf{r}}_i,$$

show that for a single electron of charge $-e$, the dipole moment operator is simply given by the position operator:

$$\hat{\mathbf{d}} = -e \hat{\mathbf{r}}.$$

(ii) Insert the completeness relation of the hydrogen eigenstates,

$$\sum_{n\ell m} |n\ell m\rangle \langle n\ell m| = \mathbb{1},$$

and show that

$$\hat{\mathbf{d}} = \sum_{n'\ell'm'} \sum_{n\ell m} \langle n'\ell'm' | \hat{\mathbf{d}} | n\ell m \rangle | n'\ell'm' \rangle \langle n\ell m|.$$

(iii) Using the completeness relation of the position eigenstates,

$$\mathbb{1} = \int d^3r |\mathbf{r}\rangle \langle \mathbf{r}|,$$

prove that dipole matrix elements can be computed from overlap integrals:

$$\langle n'\ell'm' | \hat{\mathbf{d}} | n\ell m \rangle = -e \int \psi_{n'\ell'm'}^*(\mathbf{r}) \mathbf{r} \psi_{n\ell m}(\mathbf{r}) d^3r,$$

where $\psi_{n\ell m}(\mathbf{r}) = \langle \mathbf{r} | n\ell m \rangle$.

(iv) Show that all stationary states have *no static dipole moment*, i.e.

$$\langle n\ell m | \hat{\mathbf{d}} | n\ell m \rangle = 0.$$

(Hint: use the parity of the hydrogen wavefunctions.)

(v) Compute the transition dipole matrix elements between

$$|1s\rangle \equiv |100\rangle, \quad |2s\rangle \equiv |200\rangle$$

and between

$$|1s\rangle \equiv |100\rangle, \quad |2p, m\rangle \equiv |21m\rangle \quad (m = -1, 0, 1).$$

Determine which transitions are allowed and which vanish.

Notation and useful functions

$$|1s\rangle = |100\rangle, \quad |2s\rangle = |200\rangle, \quad |2p, m\rangle = |21m\rangle.$$

$$\psi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi} a_0^{3/2}} e^{-r/a_0},$$

$$\psi_{200}(\mathbf{r}) = \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(2 - \frac{r}{a_0} \right) e^{-r/(2a_0)},$$

$$\psi_{210}(\mathbf{r}) = \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(\frac{r}{a_0} \right) e^{-r/(2a_0)} \cos \theta,$$

$$\psi_{21,\pm 1}(\mathbf{r}) = \mp \frac{1}{4\sqrt{2\pi} a_0^{3/2}} \left(\frac{r}{a_0} \right) e^{-r/(2a_0)} \sin \theta e^{\pm i\phi}.$$